Mock National Championship

12 problems · 5 hours · 1 computer per team · then a real post-mortem

Learning goals

- Run a full national-championship simulation under real conditions
- Apply Lectures 1–9 **without being told which technique to use**
- Practise triage on a set where the easy problems are **not** in order
- Produce a structured **post-mortem** that turns mistakes into action items
- Leave with an updated team notebook and a personal training plan

Format and rules

- Teams of 3, **one computer**, 5 hours, printed team notebook (≤ 25 pages)
- ICPC scoring: problems solved, then penalty = Σ (solve minute + 20 × rejected runs)
- Scoreboard freeze in the last hour
- C++17, `-O2`; stack limit = default 8 MB (beware deep recursion!)
- Statements are on the handout; the problem slides in this deck restate the tasks only

Session plan

Time	Activity
0:00 – 0:10	Setup, notebook check, template typed
0:10 – 5:10	Contest
5:10 – 5:40	Break; teams fill in the **raw timeline**
5:40 – 7:00	Debrief: problem-by-problem solutions
Next session	Team post-mortem presented (10 min/team), upsolving targets set

The problem set

	Problem	Est. rating
A	Lanterns	1300
B	Recurring Revenue	2100
C	Toll Bridges	2000
D	Distance Band	2300
E	Echo	1900
F	Corridor	1900
G	Overtime Assignment	2300
H	Twin Stations	2200
I	Xor Treasure Map	2300

Solutions

> **Solutions are not published here.** The coach holds the solutions and releases them only after the contest has been run.

- The next slides restate each task in the expected difficulty order — no ideas, no traps
- Use them in the debrief to walk through your timeline problem by problem

A — Lanterns (1300)

Task. Street [0, L]; lantern i lights $[p_i - r_i, p_i + r_i]$. Minimum number of lanterns covering all of [0, L], or −1. $n \le 2 \cdot 10^5$.

E — Echo (1900)

Task. String s ($n \le 2 \cdot 10^5$) and k. Longest substring occurring ≥ k times (overlaps allowed).

F — Corridor (1900)

Task. $n \le 2 \cdot 10^5$ integer points. Minimum width of an infinite straight strip containing them all.

C — Toll Bridges (2000)

Task. Connected multigraph, n, m, q ≤ $3 \cdot 10^5$. Query (u, v): how many edges, if removed alone, disconnect u from v?

K — Lighthouse (2000)

Task. n ≤ $2 \cdot 10^5$ ships with values at integer points ≠ origin. A beam is a closed sector of angle α (integer degrees) with apex at the origin. Max total value inside one beam position.

B — Recurring Revenue (2100)

Task. $a_i = \sum_{j=1..k} c_j\, a_{i-j} + d\cdot i + e \pmod{10^9+7}$, $k \le 20$. $q \le 2000$ queries for a_n, $n \le 10^{18}$.

H — Twin Stations (2200)

Task. n ≤ 1000 stations; station i goes at position a_i or b_i. Maximize the minimum distance between any two stations.

J — Dice Tower (2200)

Task. n dice; die i has faces with values in $[0, 10^5]$. Total faces $\le 2 \cdot 10^5$ and $S = \sum$ (max face) $\le 2 \cdot 10^5$. For each total $t \in [0, S]$, count outcomes mod 998244353.

D — Distance Band (2300)

Task. Tree, $n \le 2 \cdot 10^5$. Count unordered pairs with $L \le \text{dist}(u, v) \le R$.

G — Overtime Assignment (2300)

Task. n, m ≤ 100. Worker i can do task j for c_{ij} (or not at all). Each task done once. A worker doing t tasks pays extra $o_i \cdot t(t-1)/2$. Minimize total cost.

I — Xor Treasure Map (2300)

Task. Connected graph, weights < 2^{60}, n, m, q ≤ $2 \cdot 10^5$. Query (u, v): max xor over all walks u → v (edges reusable).

L — Warehouse Districts (2400)

Task. Positive $a_i \le 10^4$, $n \le 10^5$. Split into exactly $k \le 100$ contiguous parts minimizing Σ (part sum)^2.

Expected difficulty curve



Gold-medal pace: 9–10 problems. Advance-to-finals pace: 7–8. Solid: 5–6.

The set is **not** sorted: E, F, C, K are as easy as they look — did you find them in the first hour?

Post-mortem: process

1. **Raw timeline** (immediately): every read, submit, verdict, who was on the keyboard
2. **Per-problem table**: status, first-read time, idea time, AC time, wrong runs, root cause
3. **Root-cause categories**: reading · idea · implementation · edge case · strategy/time
4. **Counterfactual**: "with perfect play on what we *knew*, what was our score?"
5. **Three action items** max, each concrete and testable
6. **Notebook diff**: what code was missing or wrong?

Post-mortem template (per problem)

Field	Example
Problem / status	H — AC at 3:41, 2 WA
First read by / at	Alice, 0:12
Idea found at	2:50
Why so late?	Stuck on a wrong first idea; nobody re-read after 1:00
WA causes	overflow in an intermediate product; missed an edge case in the statement
Fix / action	add a self-tested template to the notebook; re-read rule at 1:30

Metrics worth tracking across mocks

- Time to first AC; time to 3 ACs
- Idle-keyboard minutes
- Wrong submissions per AC (target < 0.5)
- Problems where the idea was known but not coded in time
- "Read but never attempted" problems that turned out easy
- Upsolve completion within one week (target: every problem ≤ 2400)

Common contest-level mistakes

- Reading in letter order and missing easy E/F/K
- Two people silently working on the same problem
- Coding before writing down the idea and the complexity
- No stress test after the second WA
- Leaving the computer idle while someone thinks
- Freeze-hour panic: submitting unverified "fixes"

Practice after the mock

- Upsolve every mock problem once the coach releases the solutions
- Real sets to simulate next: ICPC North America Championship sets (Kattis), NERC finals (Codeforces Gym / contest 1912), NWERC / SWERC on Codeforces Gym

Summary

- A mock is only as valuable as its **post-mortem**
- Map each miss to a category; fix the category, not the instance
- Update the notebook the same week
- Next: World Finals track — mindset, polynomials, matching & LP, advanced trees